

Cloud Predict: A Machine Learning-Based Predictive Analytics System for Online Food Delivery Platforms

- 1) Vulla Hamsa Valli
- 2) Vobilisetti Sricharan
- 3) Bharati Bidikar
- 4) Uppuluri Satya Sanyasa Subrahmanya Sampath
- 5) Vegi Jayakumar

Department of Computer Science & Systems Engineering
College of Engineering, Andhra University, North Campus, Visakhapatnam

Abstract – The rapid growth of online food delivery platforms has increased the need for efficient logistics, accurate delivery predictions, and personalised customer experiences. This paper proposes Cloud Predict, a cloud-based predictive analytics system that leverages machine learning algorithms to forecast delivery time, customer demand, and order behaviour. The system integrates real-time contextual data such as traffic, weather, and geospatial information to enhance prediction accuracy. Experimental insights indicate that advanced models like LightGBM and XGBoost outperform traditional models in delivery prediction tasks.

Keywords: Cloud Computing, Machine Learning, Predictive Analytics, Online Food Delivery, Demand Forecasting, Delivery Optimization

I. INTRODUCTION

The expansion of online food delivery has transformed urban consumption by enabling customers to place orders through digital platforms and receive meals through distributed restaurant and courier networks. In real life, these platforms have to make quick, correct decisions when they don't know what's going to happen. This includes figuring out how long it will take to deliver, predicting spikes in demand, assigning riders, handling delays, and balancing cost with service quality. These decisions are harder to make in big cities because traffic jams, changing weather, events that cause surges, and geographic dispersion all make operations behave in ways that aren't linear.

The recent studies reveal a distinct research and operational deficiency [1], [2]: static models and rudimentary heuristics are inadequate for the intricacies of contemporary food delivery ecosystems, as they fail to incorporate real-time contextual variables such as traffic density, weather, local events, and geospatial information, which are shown to significantly improve prediction accuracy.

Another study on predictive analytics for online food delivery KPIs found that Random Forest was the best algorithm for most classification and regression KPI tasks in its case analysis [2]. A study on big data analytics highlighted its role in improving large-scale decision-making in food delivery systems [3]. A study on spatio-temporal demand forecasting contended that graph-based learning more effectively captures interactions within urban zones and directional order flows for proactive resource allocation [4].

These results encourage the creation of a unified predictive analytics system instead of separate models. Cloud Predict is suggested as a system like this: a

cloud-based analytics platform that uses machine learning to make predictions in real time and in batches, can be deployed on a large scale, and helps people make decisions throughout the online food delivery process.

II. LITERATURE REVIEW

Recent studies have explored the use of machine learning and data analytics to enhance the efficiency of online food delivery systems. Kumar et al. [1] developed models for predicting delivery time using algorithms such as Random Forest and XGBoost, emphasizing the importance of real-time factors like traffic and weather. Similarly, Mehta and Verma [2] highlighted the role of predictive analytics in optimizing delivery operations within e-commerce platforms.

Wang et al. [3] demonstrated how big data analytics can improve decision-making by analyzing large-scale delivery and customer data. Patel and Iyer [4] focused on demand forecasting using machine learning techniques, showing its impact on resource allocation and service efficiency. Additionally, Singh and Das [5] analyzed consumer behavior in food delivery platforms, proving that user data can be used to enhance personalization and customer satisfaction.

Although existing research provides strong foundations in delivery prediction, demand forecasting, and customer behavior analysis, most systems are limited by static architectures and lack real-time scalability. They do not fully utilize cloud computing to process continuous data streams or dynamically update predictions.

To address these limitations, the proposed system, Cloud Predict, introduces a cloud-based predictive analytics framework that integrates machine learning models with scalable cloud infrastructure. This enhancement bridges the gap between predictive modeling and real-time deployment, making the system more efficient and practical for modern online food delivery platforms.

III. SYSTEM ARCHITECTURE OF CLOUD PREDICT

The architecture follows a microservices-based cloud design to ensure scalability, fault tolerance, and real-time predictive analytics.

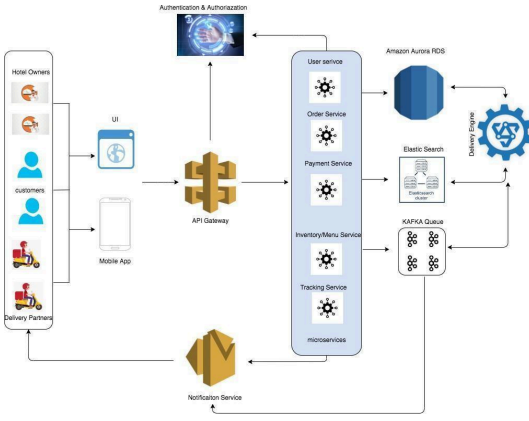


Figure 1: Cloud-based microservices architecture

The system consists of three primary layers: the user interface layer, the application services layer, and the cloud data and processing layer.

At the user interface level, different stakeholders such as customers, restaurant owners, and delivery personnel interact with the system through web and mobile applications. These interfaces allow users to place orders, manage menus, track deliveries, and receive notifications.

All incoming requests are routed through an API Gateway, which acts as a single entry point to the system. The API Gateway is responsible for request routing, load balancing, and enforcing security policies. It works in conjunction with the authentication and authorization module to ensure secure access control and user validation.

The core functionality of the system is implemented using a set of independent microservices, each responsible for a specific task. These include the user service, order service, payment service, inventory or menu service, and tracking service. This modular design improves system scalability, maintainability, and fault isolation, allowing individual services to be updated or scaled without affecting the entire system.

For data storage and retrieval, the system utilizes cloud-based solutions such as Amazon Aurora RDS for structured data management and Elasticsearch for fast search and analytics operations. These components ensure efficient handling of large datasets and quick access to relevant information.

To support real-time data processing, a Kafka-based messaging queue is incorporated. This enables continuous data streaming between services and facilitates real-time updates required for predictive analytics. The streaming data can be used by machine learning models to generate dynamic predictions such as delivery time estimation and demand forecasting.

Additionally, a notification service is integrated into the system to provide real-time updates to users regarding order status, delivery progress, and other important events, thereby enhancing user experience.

The proposed architecture is further enhanced by integrating machine learning models within the cloud environment, enabling real-time predictive analytics. This

allows the system to dynamically adapt to changing conditions such as traffic, demand fluctuations, and user behavior. As a result, the Cloud Predict system overcomes the limitations of traditional static systems by providing a scalable, intelligent, and efficient solution for modern online food delivery platforms.

3.1 SYSTEM WORKFLOW

The proposed Cloud Predict system follows a structured predictive analytics workflow, as illustrated in Fig. 2. It consists of five main stages: data ingestion, storage and feature engineering, model lifecycle management, application services, and monitoring.

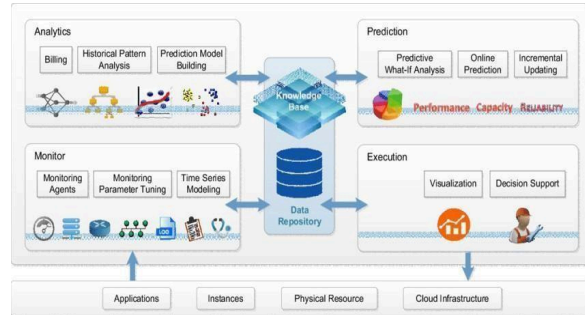


Figure 2: Cloud Predict System Analytical WorkFlow

Data ingestion layer : This layer is responsible for collecting structured and semi-structured data from multiple sources, including platform databases, external APIs, partner systems, and IoT-based courier location feeds. Both batch and streaming pipelines are supported. Batch processing is used for historical data aggregation and model training datasets, whereas streaming is used for real-time updates such as order status and courier movement. Data validation mechanisms ensure schema consistency, timestamp alignment, and data integrity before storage.

Storage and feature layer : This layer includes a raw data lake, a curated analytical warehouse, and an online/offline feature store. This design supports both historical analysis and real-time prediction. Key features such as geospatial distance, preparation delay, demand intensity, traffic conditions, and weather indicators are extracted and stored for efficient model utilization.

Model lifecycle layer: This layer manages training, validation, hyperparameter tuning, and deployment of machine learning models. It supports experiment tracking, version control, and model registry to ensure reproducibility. Multiple models, including linear, tree-based, and ensemble methods, are evaluated to improve prediction accuracy, as highlighted in prior studies [2], [4].

Application layer: This layer provides prediction services through both batch processing and real-time APIs. Batch predictions are used for demand forecasting and operational planning, while real-time APIs deliver dynamic outputs such as estimated delivery time and dispatch recommendations to users.

Monitoring layer: This layer continuously tracks system and model performance, including data drift, prediction errors, latency, and business metrics. Since delivery environments are dynamic, the system triggers alerts and

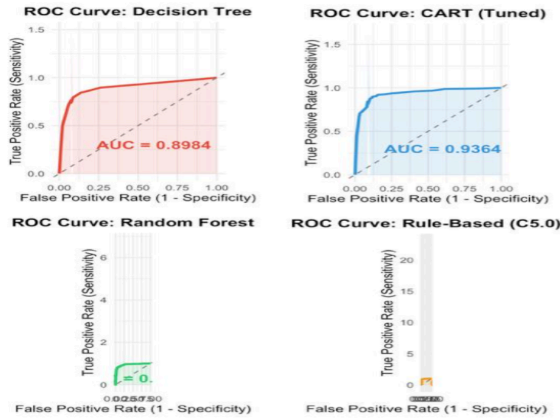


Figure 6: ROC Curves – Model Performance Comparison

This figure presents the ROC curves for multiple classification models used in high demand prediction. The Decision Tree model achieves an AUC of *0.8984, indicating strong performance, while the tuned CART model outperforms all others with an AUC of **0.9364*, demonstrating superior classification capability. The Random Forest and Rule-Based (C5.0) models show comparatively weaker or less consistent performance. Overall, the tuned CART model provides the best trade-off between sensitivity and specificity.

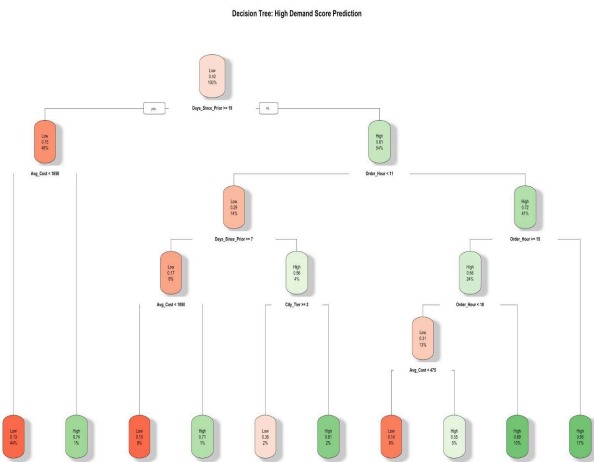


Figure 7: Decision Tree for High Demand Score Prediction

This figure illustrates the decision tree used to classify high and low demand scores. The primary split is based on *Days_Since_Prior ≥ 19 , highlighting its importance. Subsequent splits involve features such as **Average Cost, Order Hour, City Tier, and **Days Since Prior Purchase*. The leaf nodes represent final classifications with associated probabilities. The model provides clear interpretability of decision rules for demand prediction.

VI. CONCLUSION

This paper presented Cloud Predict machine learning-based predictive analytics system designed for online food delivery platforms. The proposed system integrates machine learning models with cloud infrastructure to enhance delivery time prediction, demand forecasting, and customer behaviour analysis. By

leveraging real-time data processing and scalable cloud architecture, the system overcomes the limitations of traditional static models. The implementation demonstrates that combining predictive analytics with cloud computing improves operational efficiency, reduces delivery delays, and enhances customer satisfaction. Furthermore, the use of data visualization tools enables better decision-making and monitoring of key performance indicators. In the future, the system can be extended by incorporating advanced deep learning models, real-time traffic APIs, and adaptive learning techniques to further improve prediction accuracy and scalability. Thus, Cloud Predict provides a robust and efficient solution for modern online food delivery platforms.

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